

Foreign Aid



Outline of this Lecture

- Some Stylized facts & Focus (non examinable)
- Aid, Policies, and Growth – Burnside & Dollar (2000)
- Making Progress on Foreign Aid



Stylized Facts

US – main provider of foreign aid (1960-2013)

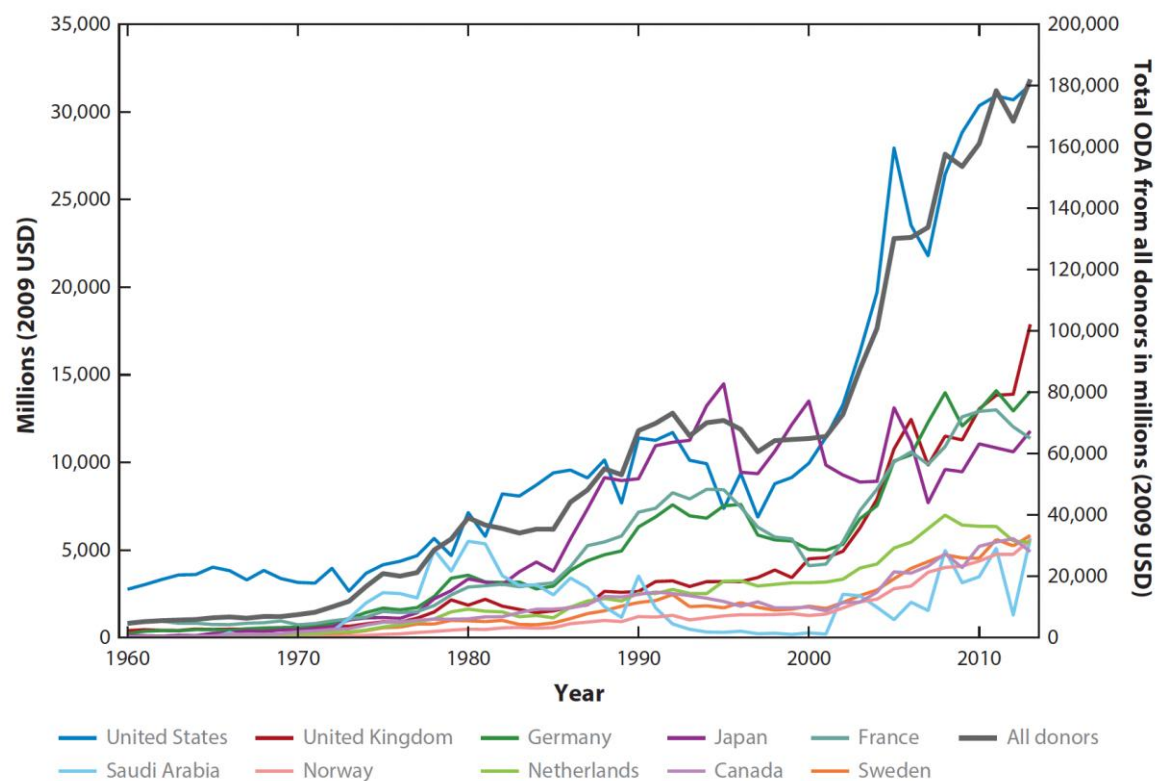


Figure 2

Total official development assistance (ODA) and ODA from the top 10 donors, 1960–2013. A top donor is defined according to ODA disbursement in 2013. Data taken from OECD table DAC 1, flows by donor/ QWIDS.

Top Aid Donors by decade

Table 1 Top donors by decade

Rank	Country	Avg. ODA	Country	Avg. ODA	Country	Avg. ODA
	1960s		1970s		1980s	
1	United States	18,311	United States	13,191	United States	15,102
2	France	4,477	Saudi Arabia	5,903	Japan	9,197
3	United Kingdom	2,399	Germany	4,777	France	6,841
4	Germany	2,283	France	4,278	Germany	6,601
5	Saudi Arabia	1,548	Japan	3,789	Saudi Arabia	6,083
6	Japan	1,155	United Kingdom	3,054	United Kingdom	3,483
7	Canada	616	Canada	2,315	Netherlands	2,947
8	Australia	600	Netherlands	1,893	Italy	2,925
9	Italy	469	Kuwait (KFAED)	1,583	Canada	2,908
10	Belgium	467	Sweden	1,478	Sweden	1,962
	1990s		2000s		2010–2013	
1	Japan	15,100	United States	22,308	United States	29,555
2	United States	13,083	Japan	11,219	United Kingdom	14,451
3	France	9,891	Germany	9,839	Germany	13,039
4	Germany	9,043	United Kingdom	9,415	France	11,559
5	United Kingdom	4,356	France	9,357	Japan	10,537
6	Netherlands	3,832	Netherlands	5,300	Netherlands	5,495
7	Italy	3,572	Spain	3,864	Sweden	5,290
8	Canada	2,941	Sweden	3,496	Canada	5,086
9	Sweden	2,559	Canada	3,429	Australia	4,835
10	Denmark	2,017	Italy	3,322	Norway	4,784

Average annual official development assistance (ODA) disbursement is reported in millions of 2009 USD. Data taken from OECD table DAC 1, flows by donor/QWIDS.

Main Recipients

Table 2 Top recipients of official development assistance (ODA) by decade

Rank	Country	Avg. ODA	Country	Avg. ODA	Country	Avg. ODA
	1960s		1970s		1980s	
1	India	5,240	Egypt	4,414	India	3,349
2	Pakistan	2,097	India	4,341	Egypt	2,838
3	Vietnam	1,662	Indonesia	4,117	Bangladesh	2,465
4	Korea	1,285	Syria	3,837	Israel	2,276
5	Algeria	1,270	Bangladesh	3,707	Indonesia	1,822
6	Brazil	1,145	Pakistan	3,614	Pakistan	1,743
7	Turkey	996	Vietnam	2,216	Syria	1,724
8	Indonesia	788	Israel	2,124	China	1,696
9	Egypt	666	Jordan	2,022	Jordan	1,526
10	Chile	566	Korea	2,019	Sudan	1,487
	1990s		2000s		2010–2013	
1	Egypt	4,198	Iraq	6,382	Afghanistan	4,850
2	China	3,552	Afghanistan	2,908	DRC	2,887
3	India	2,541	Nigeria	2,576	Vietnam	2,575
4	Indonesia	2,117	Vietnam	2,229	Ethiopia	2,508
5	Israel	2,040	Ethiopia	2,168	Pakistan	2,043
6	Bangladesh	2,031	DRC	2,018	Tanzania	2,024
7	Pakistan	1,471	Tanzania	2,009	India	1,871
8	Mozambique	1,448	Pakistan	1,922	Turkey	1,752
9	Philippines	1,420	India	1,708	West Bank and Gaza Strip	1,693
10	Tanzania	1,381	China	1,663	Kenya	1,679

Average annual ODA disbursement is reported in millions of 2009 USD. Data taken from OECD table DAC 2, flows by recipient/QWIDS.



Donor's motivations??

- Former colonial ties?
- Votes that the recipient has in the UN General Assembly
- Food shortages in the recipient countries?
- Domestic surpluses in the donor countries?

Omitted variables and reverse causality

- More aid can influence income in the recipient country not just votes in the UN
- More aid can increase ties with former colonies but also reduces “monitoring costs”
- More aid could improve infrastructure which in turn lowers the costs of delivering aid

→ We need an identification strategy

Aid Policies and Growth

Burnside – Dollar (2000)

Background - “Conditionality” at a macro - level

In the 1990s, for example, aid agencies emphasized the role of foreign aid conditional on policy

E.g., Implementation of the “Washington Consensus Reforms”: stabilization, trade liberalization and privatization (Armendáriz- Larraín 2017)

Moreover, a whole set of “institutional reforms” were advised. These reforms ranged from improved tax administration to central bank independence, better governance, transparency in the fight against corruption and more...

If and when such policies in many countries were implemented, were they conducive to higher rates of economic growth

Motivation

- Aid is an income transfer
- If it generates growth, then aid is effective
- Aid effectiveness, however, depends on how aid is used
- If it is consumed, aid is ineffective
- If it is invested, aid is effective
- Effectiveness of aid conditional on “**policies**”

Empirical investigation

$$g_{it} = \alpha_t + \beta \text{Aid/GDP}_{it} + \mathbf{X}'\Gamma + \varepsilon_{it}$$

- i denotes a country and t one of six a 4-year time periods from 1970 to 1993.
- Aid/GDP_{it} is a country's average aid-to-GDP ratio during the 4-year period t .
- Controls include log of initial per capita GDP.

TABLE 3—GROWTH REGRESSIONS: USING ALL COUNTRIES
AND THE INDIVIDUAL POLICY VARIABLES

Estimation method	(1)	(2)	
	OLS	OLS	2SLS
Initial GDP	−0.65 (0.55)	−0.61 (0.58)	−0.74 (0.62)
Ethnic fractionalization	−0.58 (0.73)	−0.53 (0.73)	−0.69 (0.78)
Assassinations	−0.44* (0.27)	−0.44* (0.27)	−0.44 (0.27)
Ethnic fractionalization × assassinations	0.81* (0.45)	0.81* (0.45)	0.81* (0.46)
Institutional quality	0.64** (0.17)	0.64** (0.17)	0.63** (0.17)
M2/GDP (lagged)	0.015 (0.015)	0.014 (0.015)	0.017 (0.016)
Sub-Saharan Africa	−1.53** (0.73)	−1.61** (0.76)	−1.35* (0.76)
East Asia	0.89 (0.56)	0.93* (0.57)	0.80 (0.58)
Budget surplus	6.85** (3.39)	7.00** (3.38)	6.49* (3.47)
Inflation	−1.40** (0.41)	−1.40** (0.41)	−1.39** (0.41)
Openness	2.16** (0.51)	2.12** (0.50)	2.25** (0.54)
Aid/GDP	—	0.036 (0.13)	−0.085 (0.19)

Interpreting the Results

Concerns with these results?

- Endogeneity of the aid-to-GDP ratio: reverse causality and omitted variables bias.
- The impact of aid on economic growth may depend on what is done with the foreign aid.

Foreign Aid (Cont..)

Equation

$$g_{it} = \alpha_t + \beta_1 Aid/GDP_{it} + \beta_2 Aid/GDP_{it} \times Policy_{it} + \beta_3 Policy_{it} + \mathbf{X}'\Gamma + \varepsilon_{it}$$

- $Policy_{it}$ is a policy index created by the authors.
- The Aid/GDP_{it} variable, and its interaction, is instrumented using 2SLS.

TABLE 1—SUMMARY OF REGRESSION SPECIFICATIONS AND IDENTIFICATION

Variable	Equation		
	Variants of (4), growth		
Endogenous variables			
Real growth rate	LHS	LHS	LHS
Aid/GDP	RHS	RHS	RHS
$(\text{Aid/GDP}) \times \text{policy}$		RHS	RHS
$(\text{Aid/GDP})^2 \times \text{policy}$			RHS
Exogenous variables			
Logarithm of initial income	Included	Included	Included
Policy index	Included	Included	Included
Institutional quality	Included	Included	Included
Ethnic fractionalization	Included	Included	Included
Assassinations	Included	Included	Included
Ethnic fractionalization $\times$ assassinations	Included	Included	Included
M2/GDP, lagged	Included	Included	Included
Logarithm of population			
Arms imports/imports, lagged			
Sub-Saharan Africa dummy	Included	Included	Included
East Asia dummy	Included	Included	Included
Egypt dummy			
Franc zone dummy			
Central America dummy			
Logarithm of initial income $\times$ policy			
Logarithm of population $\times$ policy			
Arms imports/imports, lagged $\times$ policy			
$(\text{Logarithm of initial income})^2 \times \text{policy}$			
$(\text{Logarithm of population})^2 \times \text{policy}$			

Notes: LHS indicates that a variable is included as the left-hand-side variable. RHS indicates that a variable is a right-hand-side variable. All exogenous variables are used as instruments in 2SLS estimation.

Constructing the policy index

The policy index is a weighted average of:

1. Indicator for “trade openness” from Sachs and Warner (1995), which is itself a weighted average of its own components: BMP, tariffs, and government intervention in exports.
2. Inflation: evidence of poor monetary policy.
3. Budget surplus.

Weights are chosen by regressing per capita GDP on the variables, and then using the coefficients.

TABLE 3—GROWTH REGRESSIONS: USING ALL COUNTRIES
AND THE INDIVIDUAL POLICY VARIABLES

	(1)	(2)	
Estimation method	OLS	OLS	2SLS
Initial GDP	−0.65 (0.55)	−0.61 (0.58)	−0.74 (0.62)
Ethnic fractionalization	−0.58 (0.73)	−0.53 (0.73)	−0.69 (0.78)
Assassinations	−0.44* (0.27)	−0.44* (0.27)	−0.44 (0.27)
Ethnic fractionalization × assassinations	0.81* (0.45)	0.81* (0.45)	0.81* (0.46)
Institutional quality	0.64** (0.17)	0.64** (0.17)	0.63** (0.17)
M2/GDP (lagged)	0.015 (0.015)	0.014 (0.015)	0.017 (0.016)
Sub-Saharan Africa	−1.53** (0.73)	−1.61** (0.76)	−1.35* (0.76)
East Asia	0.89 (0.56)	0.93* (0.57)	0.80 (0.58)
Budget surplus	6.85** (3.39)	7.00** (3.38)	6.49* (3.47)
Inflation	−1.40** (0.41)	−1.40** (0.41)	−1.39** (0.41)
Openness	2.16** (0.51)	2.12** (0.50)	2.25** (0.54)
Aid/GDP	—	0.036 (0.13)	−0.085 (0.19)
Partial R^2 of first-stage regressions			
Aid/GDP	—	—	0.44
Test for exogeneity of the aid variables			
$\chi^2(1)$	—	—	0.61 [0.43]
Other statistics			
Observations	275	275	275
$\bar{R}^2$	0.35	0.35	0.35

Policy index →

The authors then use this regression to create a **policy index** comprised of (a) **budget surplus**, (b) **inflation**, and (c) **trade openness**

$$\text{Policy} = 1.28 + 6.85 \times \text{Budget surplus} \\ - 1.40 \times \text{Inflation} + 2.16 \times \text{Openness}.$$

TABLE 3—GROWTH REGRESSIONS: USING ALL COUNTRIES
AND THE INDIVIDUAL POLICY VARIABLES

Estimation method	(1)	(2)	
	OLS	OLS	2SLS
Initial GDP	−0.65 (0.55)	−0.61 (0.58)	−0.74 (0.62)
Ethnic fractionalization	−0.58 (0.73)	−0.53 (0.73)	−0.69 (0.78)
Assassinations	−0.44* (0.27)	−0.44* (0.27)	−0.44 (0.27)
Ethnic fractionalization × assassinations	0.81* (0.45)	0.81* (0.45)	0.81* (0.46)
Institutional quality	0.64** (0.17)	0.64** (0.17)	0.63** (0.17)
M2/GDP (lagged)	0.015 (0.015)	0.014 (0.015)	0.017 (0.016)
Sub-Saharan Africa	−1.53** (0.73)	−1.61** (0.76)	−1.35* (0.76)
East Asia	0.89 (0.56)	0.93* (0.57)	0.80 (0.58)
Budget surplus	6.85** (3.39)	7.00** (3.38)	6.49* (3.47)
Inflation	−1.40** (0.41)	−1.40** (0.41)	−1.39** (0.41)
Openness	2.16** (0.51)	2.12** (0.50)	2.25** (0.54)
Aid/GDP	—	0.036 (0.13)	−0.085 (0.19)
Partial R^2 of first-stage regressions			
Aid/GDP	—	—	0.44
Test for exogeneity of the aid variables			
$\chi^2(1)$	—	—	0.61 [0.43]
Other statistics			
Observations	275	275	275
$\bar{R}^2$	0.35	0.35	0.35

TABLE 4—GROWTH REGRESSIONS: USING ALL COUNTRIES AND THE POLICY INDEX

Estimation method	(3)		(4)		(5)	
	OLS	2SLS	OLS	2SLS	OLS	2SLS
Initial GDP	−0.61 (0.56)	−0.79 (0.59)	−0.56 (0.56)	−0.71 (0.60)	−0.60 (0.57)	−0.90 (0.65)
Ethnic fractionalization	−0.54 (0.72)	−0.70 (0.75)	−0.42 (0.73)	−0.47 (0.83)	−0.42 (0.72)	−0.73 (0.81)
Assassinations	−0.44* (0.26)	−0.43 (0.27)	−0.45* (0.26)	−0.44* (0.26)	−0.45* (0.26)	−0.41 (0.27)
Ethnic fractionalization × assassinations	0.82* (0.44)	0.78* (0.44)	0.80* (0.44)	0.75* (0.45)	0.79* (0.44)	0.71 (0.45)
Institutional quality	0.64** (0.17)	0.63** (0.17)	0.67** (0.17)	0.68** (0.19)	0.69** (0.17)	0.66* (0.18)
M2/GDP (lagged)	0.014 (0.013)	0.019 (0.015)	0.016 (0.014)	0.025 (0.017)	0.012 (0.014)	0.017 (0.016)
Sub-Saharan Africa	−1.60** (0.73)	−1.31* (0.72)	−1.84** (0.74)	−1.71** (0.82)	−1.87** (0.75)	−1.29 (0.84)
East Asia	0.91* (0.54)	0.81 (0.53)	1.20** (0.58)	1.27** (0.63)	1.31** (0.58)	1.15* (0.56)
Policy index	1.00** (0.14)	1.01** (0.14)	0.78** (0.20)	0.65** (0.30)	0.71** (0.19)	0.74* (0.20)
Aid/GDP	0.034 (0.12)	−0.12 (0.18)	0.49 (0.12)	−0.10 (0.21)	−0.021 (0.16)	−0.32 (0.36)
(Aid/GDP) × policy	—	—	0.20** (0.09)	0.37 (0.33)	0.19** (0.07)	0.18* (0.10)
(Aid/GDP) ² × policy	—	—	−0.019** (0.0084)	−0.038 (0.038)	—	—
Partial R ² of first-stage regressions						
Aid/GDP	—	0.44	—	0.42	—	0.29
(Aid/GDP) × policy	—	—	—	0.16	—	0.60
(Aid/GDP) ² × policy	—	—	—	0.11	—	—
Test for exogeneity of the aid variables						
$\chi^2(j)$	—	1.10 [0.29]	—	0.85 [0.84]	—	1.51 [0.47]
Other statistics						
Observations	275	275	275	275	270	270
$\bar{R}^2$	0.36	0.35	0.36	0.34	0.36	0.35

Conclusion B&D (2000)

- Aid has a positive impact on GDP per capita
- Provided developing countries (aid recipients) have good trade, monetary and fiscal policies
- Aid should be more effective if it were more systematically granted on condition of good policy



Making Progress on Foreign Aid

Qian (2015)



Main Arguments

- Foreign aid to very poor countries accounts for very little of total global aid
- Foreign aid is often determined by the objectives of the donors not by the recipient countries' need
- Measurement and identification problems
- Recent studies

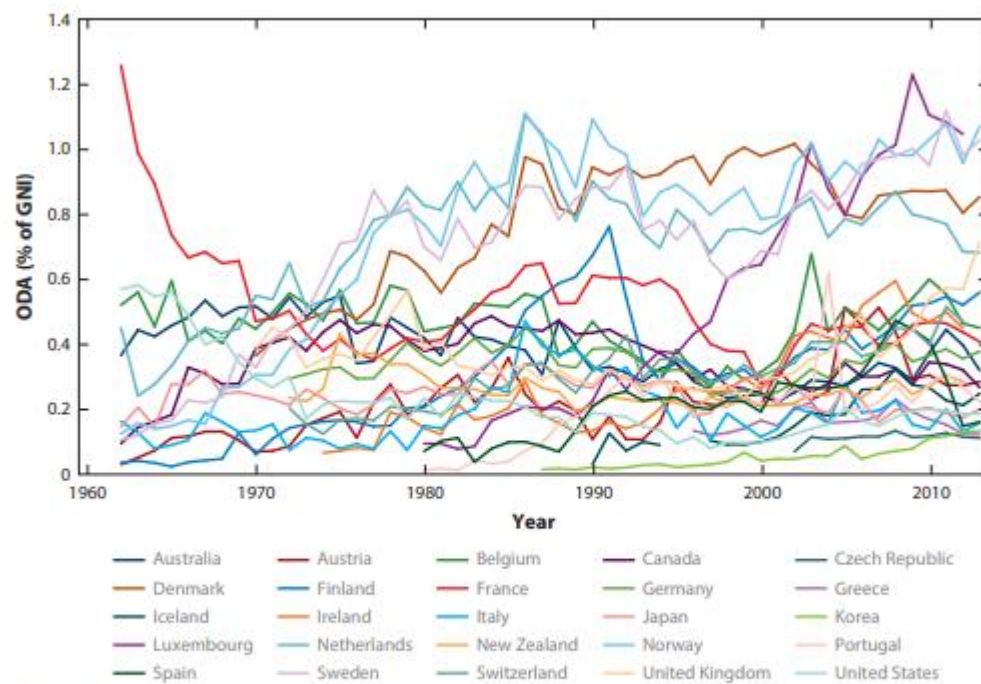


Figure 1

Official development assistance (ODA) as a percentage of gross national income (GNI), 1960–2013. Data taken from OECD table DAC 1, flows by donor/QWIDS, and from WDI for GNI.



How can foreign aid matter?

- Growth
- Conflict
- Political Accountability

Table 5 Correlates of official development assistance (ODA)

	Dependent variable: Ln ODA			
	(1)	(2)	(3)	(4)
Ln population	−0.498 (0.00107)	−0.952 (0.000)	−1.638 (0.0325)	−1.824 (0.0319)
Lag log GDP	1.372 (0.000)	1.866 (0.000)	1.423 (0.000)	1.718 (0.000)
Lag growth	0.00345 (0.932)	−0.0226 (0.618)	0.0112 (0.497)	0.0264 (0.257)
Lag conflict incidence	0.0273 (0.164)	0.0172 (0.280)	0.00222 (0.786)	0.000203 (0.982)
Small natural disaster	−0.0486 (0.254)	−0.0843 (0.129)	0.0212 (0.0628)	0.0255 (0.0583)
Large natural disaster	0.00963 (0.507)	0.0162 (0.386)	−0.00398 (0.636)	−0.00951 (0.397)
Controls				
Year fixed effects	No	Yes	No	Yes
Country fixed effects	No	No	Yes	Yes
Observations	1,081	1,081	1,081	1,081
R ²	0.85	0.89	0.89	0.95

The Consequences of Foreign Aid

- Growth on conditional aid (Burnside – Dollar, 2000)
- Mitigation of negative export shocks (Collier & Dehn (2001)
- Galiani et al (2014)
- Easterly (2004), Roodman (2007)
- Rajan & Subramanian (2011)

Positive

Negative

The Consequences of Foreign Aid (Cont..)

- Food Aid driven by production shocks in the US, but aid causes conflict (Nunn & Qian (2014))
- Aid causes conflict (Cross et al 2014)
- Increased political accountability (Eubank 2012)
- A large number of context – specific RCTs some of which we have studied already

Methodological Issues and Measurement errors

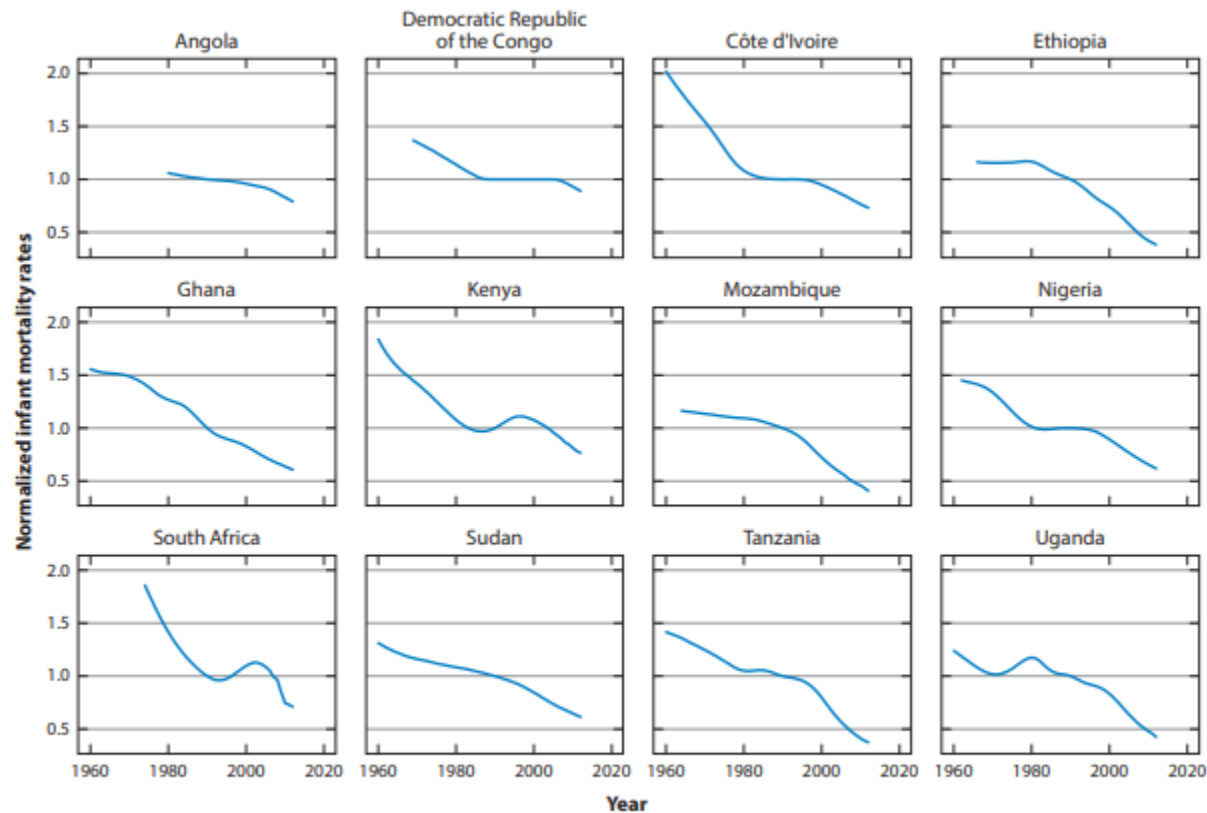


Figure 12

Infant mortality rates in sub-Saharan Africa. The values for each country are normalized by dividing by the 1990 value for that country. Data taken from WDI.

Takeaways

- Main aid donor has been the US but, more recent evidence suggests that China is not too far apart
- Main recipients of aid – less clear (not always the ones that need it the most, i.e., often leads to conflict).
- Suspicion on not delivered on the basis of “need” is widespread among political scientists and economists
- Are aid flows “strategic”?

Takeaways (cont..)

- Economists have attempted to go beyond “correlations”
- Burnside and Dollar (2000) contend that there is a causal effect of aid on economic growth
- When recipients have “Good Policies”/ “Washington Consensus Policies”, foreign aid has an effect on growth and therefore is effective
- Their results have been questioned by Easterly (a leading contrarian to the provision of aid) because aid can potentially have negative effects in the recipient countries

Takeaways (Cont..)

- More recent research focuses on food aid by the US (Nunn and Qian (2014))
- The recipients' needs are better served when US wheat production is high
- Recipients' political cycles can potentially explain donors' aid amounts.
- In turn focus of attention on low and income countries democracies
- Foreign aid is potentially strengthening “democratic institutions” which in turn brings us back to the ideas discussed in the last lecture.
- Can institutions change?
- If so, foreign aid delivered by OECD democracies might be playing a role on strengthening institutions of private property / or more inclusive .

2 Reminders

- **Research paper** (candidate number(s) only) are due **this Friday Friday 28th March 2025 at 17h00 or 5 PM** – London time via the Turnitin link already on Moodle.
- My office hour is tomorrow and next Tuesday 12h – 14h. Research topics and progress cannot be discussed any longer.

Thank you 😊